

48V Electrical Architecture

Document role

- This file owns the final 48V electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in docs/implementation/ .
- Keep unresolved vendor gates, risk state, and follow-up closure items in docs/core/TRACKING.md .
- Keep broad project sequencing and day-to-day execution framing in docs/core/PROJECT.md or the active plan docs.

As-of date: 2026-08-12

Purpose: hold the finalized, concise 48V house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- [Electrical topology diagram](#)
- [Electrical fuse schedule](#)
- [Mechman / WS500 / APM-48 install guide](#)
- [Systems](#)
- [Tracking](#)
- [Estimated BOM](#)

Final direction

- House architecture stays 48V core with 3x 51.2V 100Ah batteries in parallel.
- Alternator charging is the dedicated 48V secondary alternator path; obsolete pre-Mechman charger hardware is removed from active planning.
- Protection baseline is Mechman + WS500 + Balmar APM-48 .
- Manual alternator-charge enable/disable is through Ford Upfitter Switch #3 .
- Upfitter #3 is a low-current control signal only. It does not carry alternator output current.
- WS500 white Feature-In is reserved for future fault-interlock work, not required in Phase 1.

Current commissioning state (2026-09-02)

- 48V bus has been live-tested: owner measured 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II DC/inverter mode has been switched on with inverter light illuminated, slight normal hum, and no reported error lights.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect.
- Cerbo GX access point/remote-console workflow is active; Cerbo power is a small inline fused feed from the 48V system side and MultiPlus communication is via VE.Bus RJ45.
- Shore charging has been tested through the MultiPlus at household-outlet current limits. The first short test proved basic AC-in/charger function; later owner verification redid the settings and confirmed first-battery behavior entered bulk, then quickly transitioned to absorption at/near 100% as planned.
- MultiPlus LiFePO4 charge-profile programming/verification is treated as closed for the current shore-charger setup. Current target is 56.8V absorption/charge, 54.0V float, short absorption dwell, equalization off, conservative LiFePO4 storage behavior, and source-current limit matched to the actual shore circuit. The Dumfume manual's 58.4V +/-0.2V value is documented, but because the same manual also lists 58.4V as charge-limit/over-charge protection voltage, it is not the active routine charger target. DVCC remains disabled unless a documented BMS/GX control path is added.
- The electrical module is hard-mounted and tied into the hard-mounted Bench/Galley structure. Owner reports the batteries and ICECO are positively restrained; the not-yet-installed air fryer, bench lid, monitor/laptop package, and other storage remain separate retention work.
- All three batteries' 2/0 AWG branch leads are complete, and the 3x 48V bank has been permanently paralleled and in regular use for an extended period. The earlier Battery 2/3 isolated-charge staging language is historical and no longer an active prerequisite.
- Owner reports regular use of the 12V , 48V , and 120V systems. All four duplex outlet boxes and the other 120V enclosures are complete, and the owner confirms the MultiPlus is grounded to the truck chassis. Remaining AC safety verification is explicit inspection/continuity proof of the installed MultiPlus M6 PE bond, the separate conductive-shell bond, GFCI LINE/LOAD /downstream trip behavior, polarity, PE continuity, and neutral-isolation behavior—not another enclosure or chassis-ground build.
- Solar pass-through, disconnect, fixed wiring, and MPPT landing are complete; no panels are connected. The secondary alternator remains disconnected, with the protected PH-VAN red lead and Wakespeed read/save/programming preceding controlled first charge.

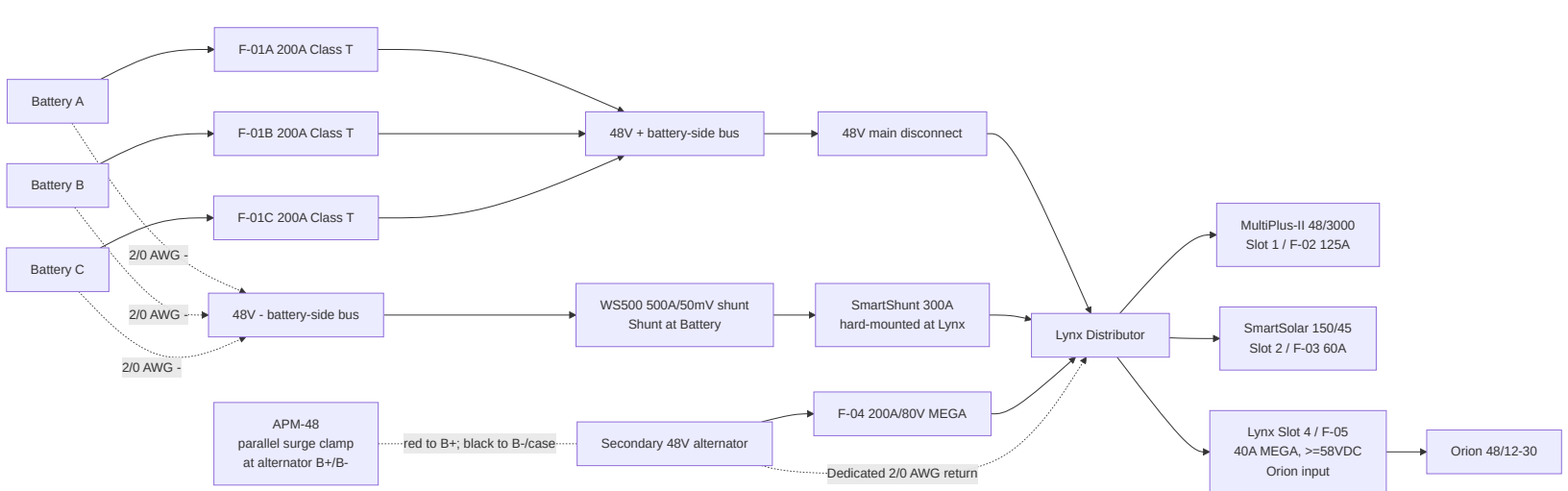
Locked component set

Function	Locked baseline	BOM row(s)
House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6
Battery branch protection	F-01A/B/C 200A Class T; manual-backed battery limit is 200A max continuous discharge per battery, with terminal/manufacturer fuse guidance still not separately specified; owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30; fed directly from Lynx Slot 4 through one verified 40A MEGA (58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) into the existing 6 AWG input pair; no separate inline/DIN input fuse; keep 6 AWG and F-07 60A/80V on 12V output	20, 10, 11
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Solar controller	Purchased Victron SmartSolar MPPT 150/45; retain it for the final array	25
Solar array	Purchased 4x Renogy 175W flexible monocrystalline = 700W; current candidate is one 4S string at 78.0V Vmp, 95.6V Voc, 8.98A Imp, and 9.50A Isc. Published -0.31%/C Voc coefficient gives 114.86V nominal calculated Voc at -40C. Exact received labels and hot-start/tracking behavior remain acceptance gates.	24
Alternator kit	Mechman 48V secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 200A/80V MEGA in Lynx Slot 3	170
WS500 PH-VAN and shunt sensing	One Eaton/Bussmann HEB-AA in-line holder + Littelfuse KLKD015.T (15A, 600VAC/DC) midget fuse on the short red combined regulator-power/positive-sense lead; separate 500A/50mV Wakespeed shunt on the battery side of the hard-mounted Victron SmartShunt in the common battery-negative path	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

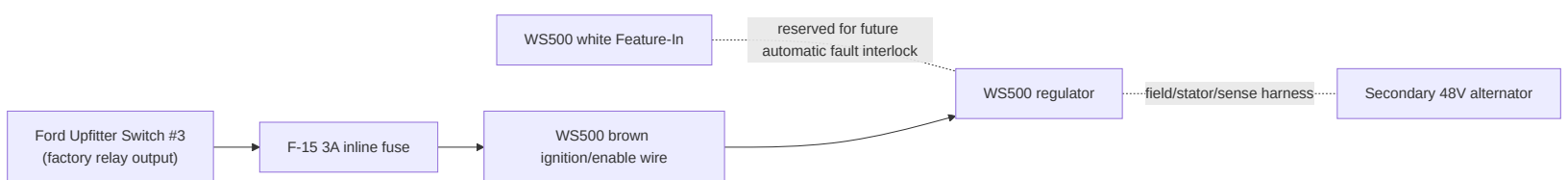
Battery manual limits and charger-programming baseline

- Battery model basis: Dumfume 51.2V 100Ah LiFePO4, 3x in parallel (1S3P ; manual allows up to 1S4P).
- Per-battery charge voltage: 58.4V +/-0.2V ; charge-limit/over-charge protection voltage also listed as 58.4V .
- Per-battery current references: recommended charge 20A ; max continuous charge 100A ; recommended discharge 50A ; max continuous discharge 200A .
- Bank-level reference for current 3x setup: recommended charge 60A ; max continuous charge 300A ; recommended discharge 150A ; max continuous discharge 600A .
- Protection thresholds: over-discharge protection 36.8V , recovery 43.2V ; discharge overcurrent protection 600A ; short-circuit protection 1800A .
- Temperature thresholds: low-temp charge protection approx 41F-50F with recovery at 50F ; high-temp protection listed as 157F / 140F .
- MultiPlus-II charger limit is 35A , so full-output charging is below the bank's 60A recommended-charge reference (~11.7A per battery in 3P). For first household-outlet tests, keep AC input current at 10A , then 12A max on a normal 15A source. Current charger-programming target is 56.8V absorption/charge, 54.0V float, 52.8V storage if shown, 0.5h max absorption, equalization off, and temperature compensation off.

48V power path



Alternator control path



Control logic

- Upfitter #3 ON : WS500 brown wire is energized, regulator is allowed to run, alternator charging can occur.
- Upfitter #3 OFF : WS500 is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use Upfitter #3 as the manual alternator-charge shutdown control.
- Do not use the main 48V disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why Upfitter #3

- It is already a factory-switched, relay-driven 12V control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
F-01A/B/C	Battery branch positive protection	200A Class T provisional	2/0 AWG
F-04	Alternator branch into Lynx Slot 3	200A/80V MEGA	2/0 AWG
F-05	Lynx Slot 4 -> Orion 48V input	MEGA 40A, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback	Existing 6 AWG direct to Orion; single input fuse
F-06	Retired standalone Orion input fuse position	Not installed	MIDI/FKS/DIN concepts superseded; do not stack with F-05
F-12/F-13-PHVAN	WS500 PH-VAN combined regulator-power / positive-voltage-sense red lead	Littelfuse KLKD015.T, 15A fast-acting midget fuse, 600VAC/DC, in Eaton/Bussmann HEB-AA 600V in-line holder	In-line beside F-04 load/alternator-side stud; short 14 AWG pigtails; sealed step-down splice to 16 AWG harness; no panel, DIN rail, or enclosure

CERB0-PWR	Cerbo GX low-current power feed	1A-3A inline fuse/holder rated for the 48v bank maximum; system/load side of main disconnect preferred for bench shutdown	18 AWG red/black duplex acceptable
WS500 current-sense pair	Purple/grey to Wakespeed shunt in common battery-negative path	No fuse; purple/high to battery-negative side, grey/low to SmartShunt/Lynx system side; twist pair and route away from noise; configure Shunt at Battery	harness lead
F-15	Upfitter #3 to WS500 brown ignition wire	3A inline ATO/ATC on 12V control circuit	16 AWG TXL/GXL control wire

Major conductors

- Battery branch and main 48V trunk wiring: 2/0 AWG .
- Parallel battery-current sharing target is similar **total loop resistance** per battery path, not equal positive-only length. The current bench layout may use short/medium/long positive leads balanced by long/medium/short negative leads; do not add unnecessary cable coils solely to make positive leads identical.
- Secondary alternator positive path (ALT B+ -> F-04 -> Lynx): 2/0 AWG , ~20 ft one-way planning basis. The APM-48 is wired in parallel across alternator B+ and B- /case; it is not in series with this conductor.
- Secondary alternator dedicated negative return (ALT B-/case -> long 2/0 -> Lynx -): 2/0 AWG , ~20 ft one-way planning basis. Do not use sheet metal/chassis as the normal return.
- Common battery-negative measurement path: battery negative combine -> Wakespeed 500A/50mV shunt -> Victron SmartShunt -> Lynx/system negative . The SmartShunt remains hard-attached to the Lynx. Configure Shunt at Battery ; purple/high faces battery negative and grey/low faces the SmartShunt/Lynx system side.
- Orion 48V feeder: existing 6 AWG remains as the no-rework path from Lynx Slot 4 and is protected by F-05 40A MEGA (>=58VDC); it is electrically overkill, and flexible fine-strand 10 AWG would be adequate if ever replaced for unrelated reasons. MPPT battery leads and Orion 12V output remain 6 AWG .
- Upfitter #3 control lead to WS500 brown wire: 16 AWG TXL/GXL planning basis, ~6 ft one-way assumed until measured.

Grounding and bonding separation

- **AC protective earth:** the MultiPlus external M6 PE lug is mandatory in this mobile installation. Owner report 2026-09-02 : the MultiPlus is grounded to truck chassis, closing physical installation of that bond. At the next de-energized inspection, verify the connection originates at the M6 PE lug, uses at least the Victron-manual 4 mm² conductor (current build default 10 AWG green stranded copper), and lands at a sound truck-chassis bond point. Do not use the aluminum camper shell or 80/20 as the only protective-earth conductor.
- **Exposed aluminum shell:** add a separate corrosion-compatible bonding jumper from the Hiatus shell to the chassis/equipment-ground network and verify low-resistance continuity. Mechanical shell/bed mounting contact is not accepted as proof of a durable bond.
- **AC neutral:** do not add a fixed neutral-ground bond downstream. The MultiPlus internal ground relay remains enabled: AC-out neutral bonds to chassis in inverter mode and that bond opens when external AC is accepted.
- **DC negatives:** do not jumper MultiPlus PE/case to Lynx negative or the 12V negative bus. Keep normal 48V / 12V current on dedicated returns. The Mechman branch uses a dedicated 2/0 negative directly to Lynx/system negative. Both monitoring shunts remain in series in the one common battery-negative path, so all battery current must pass through battery -> Wakespeed shunt -> SmartShunt -> Lynx/system regardless of alternator case/chassis behavior. The SmartShunt remains directly attached to the Lynx; prove no return is landed on the battery side of either shunt before charging.

APM-48 wiring intent

- Mount the APM-48 at the alternator end of the 48V branch, as close to the alternator output/ground points as practical.
- Treat the APM-48 as a parallel surge/load-dump clamp across the alternator, not a series device in the alternator charge-current path.
- Connect APM red to alternator B+ .
- Connect APM black to alternator B- , or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do **not** place either APM connector under the main battery cable lugs; Balmar's quick-start sheet explicitly prohibits placing APM connectors under the battery cable lugs.

Normal operating sequence

1. Main 48V disconnect closed.
2. Engine running.
3. Upfitter #3 switched ON .
4. WS500 enabled and regulating.
5. Alternator charges the house bank through APM-48 and F-04 .

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch Upfitter #3 OFF to disable the WS500 .
2. Wait for alternator charge current to collapse.
3. Open the main 48V disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The APM-48 is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the 48V bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- F-04 200A/80V MEGA stock and the Eaton/Bussmann HEB-AA + Littelfuse KLKD015.T in-line PH-VAN branch package were ordered 2026-08-09 ; receipt, body-marking, physical fit, and small termination-stock checks remain open. Procurement/physical inventory confirmation for the separate F-15 3A Upfitter-control hardware also remains open.
- Physical fit, stud-size, and jumper-length confirmation for the Wakespeed shunt inserted on the battery side of the SmartShunt in the common battery-negative path; the SmartShunt-to-Lynx connection does not move.
- Exact Mechman field-voltage/WS500 derate setting and alternator negative/case-isolation behavior in the installed kit.
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the WS500 .
- Whether future automatic fault-interlock logic should be added on the WS500 Feature-In or through an external relay.
- Solar purchase release: final MaxxAir/track survey, 1:1 panel/Starlink templates, two-panel cassette design, preliminary itemized <=75 lb budget, and any required insurer/Hiatus acceptance. Product/topology are locked; purchase is not yet released.
- Solar installation/travel release: received labels/polarity, built-cassette load/uplift proof, actual complete <=75 lb weigh-in, branch/disconnect/cable/umbilical/entry hardware, bond-coupon/removal proof, estimated hot-string voltage minus measured route drop, and stationary hot-roof commissioning.

Rule for future edits

- Update this file first for any 48V architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.